

What is Entropy?

- Entropy is a measure of disorder or randomness in a system.
- It indicates how energy is spread out.

Basic Description

- Entropy change: $\Delta S = \int (dQ/T)$.
- Entropy increases in spontaneous processes.
- It is a state function.

Entropy in Thermodynamics

- Second Law: Total entropy of an isolated system never decreases.
- Spontaneous processes increase entropy.
- Examples: mixing, diffusion.

Use Cases of Entropy

- Use cases:
 - • Heat engines
 - • Predicting direction of processes
 - • Understanding efficiency
 - • Gas expansions and thermal equilibrium

Entropy in Nature and the Universe

Stars burning fuel (entropy increases)

Expansion of the universe

Heat death concept (very basic explanation)

Conclusion

- Entropy helps describe natural processes.
- It shows why some processes happen naturally.
- It is fundamental in thermodynamics.